<table>
<tr><td>Course Code:
CSE 2010</td><td colspan="2">Course Title: Operating Systems
Type of Course: Theory Only</td><td>L- P- C</td><td>3</td><td>0</td><td>3</td></tr>
<tr><td>Version No.</td><td colspan="6">2.0</td></tr>
<tr><td>Course Pre-requisites</td><td colspan="6">Basic knowledge on computers, computer software & hardware, and Computer Organization.</td></tr>
<tr><td>Anti-requisites</td><td colspan="6">Nil</td></tr>
<tr><td>Course Description</td><td colspan="6">Operating systems being central to computing activities, this Course provide understanding of the functions and functional modules of operating systems. The design and implementation of Operating systems is also covered.</td></tr>
<tr><td>Course Objective</td><td colspan="6">The objective of the course is to familiarize the learners with the concepts of Operating Systems and attain SKILL DEVELOPMENT through PARTICIPATIVE LEARNING techniques</td></tr>
<tr><td>Course Out Comes</td><td colspan="6">On successful completion of the course the students shall be able to:
CO1: Describe the fundamental concepts of operating Systems [Knowledge Level]
CO2: Demonstrate various CPU scheduling algorithms. [Application Level]
CO3: Apply synchronization tools to a given problem. [Application Level]
CO4: Discuss various memory management techniques.[Comprehension Level]</td></tr>
<tr><td>Course Content:</td><td colspan="6"></td></tr>
<tr><td>Module 1</td><td>Introduction</td><td>Assignment</td><td colspan="3">Data Analysis task</td><td>7 Sessions</td></tr>
<tr><td colspan="7">Topics: Overview of OS and design, Introduction- Computer System Architecture, Operating System Structure, Operations, Computing environments, OS implementation, Operating System Services, User and OS interface, System Calls and its types, System Programs [loaders, linkers], UNIX/LINUX commands: System Programs[CLI/SHELL, loaders, linkers]</td></tr>
<tr><td>Module 2</td><td>Process Management</td><td>Assignments</td><td colspan="3">Analysis, Data Collection</td><td>10 Sessions</td></tr>
<tr><td colspan="7">Topics: Process Concept, Operations on Processes, Inter Process Communication, Introduction to threads - Multithreading Models, Process Scheduling– Basic concepts, Scheduling Criteria, Scheduling Algorithms: FCFS, SJF, RR, Priority, Multilevel Queue, Linux Scheduler, CASE STUDY: Linux Scheduler</td></tr>
<tr><td>Module 3</td><td>Process Synchronization and Deadlocks</td><td>Quiz</td><td colspan="3">Case studies / Case let</td><td>10 Sessions</td></tr>
<tr><td colspan="7">Topics: The Critical-Section Problem- Peterson’s Solution, Synchronization hardware, Test and Set, Mutex locks, Semaphores, Advanced Synchronization Problems-IBM Quality and implementation, Monitors. Introduction to Deadlocks, Deadlock Characterization, Methods for handling deadlock: Deadlock Prevention and Implementation, Deadlock Avoidance and Implementation Deadlock detection & Recovery from Deadlock.</td></tr>
<tr><td>Module 4</td><td>Memory Management and File Systems</td><td>Assignment</td><td colspan="3">Case Studies / Case let</td><td>11 Sessions</td></tr>
</table>

Topics: Introduction to Memory Management, Swapping, Contiguous and Non-Contiguous Memory Allocation, Segmentation, Paging - Structure of the Page Table – Demand Paging – Page Replacement, Allocation of Frames – Thrashing. RAID Structures: Disk Scheduling, RAID LEVELS

Targeted Application & Tools that can be used: UNIX

Project work/Assignment:

Mini Project: Demonstration of File Handling techniques/Memory and Disk Management.

Text Book

T1: Silberschatz A, Galvin P B and Gagne G, “Operating System Concepts”, 9th edition Wiley, 2013.

References

R1. William Stallings, “Operating systems”, Prentice Hall, 7th Edition, Pearson, 2013.

R2. Andrew S Tanenbaum and Albert S Woodhull, “Operating Systems Design and Implementation”, 3rd Edition, Pearson, 2015.

E book link R1: Details for: Operating systems : internals and design principles › Koha online catalog

E book link R2: Details for: Operating systems : design and implementation › Koha online catalog

R3 Web resources:

1)https://www.youtube.com/watch?v=vBURTt97EkA&list=PLBlnK6fEyqRiVhbXDGLXDk_OQAeuVcp2O

2)https://www.youtube.com/watch?v=3-ITLMMeeXY&list=PL3pGy4HtqwD0n7bQfHjPnsWzkeR-n6mkO

3)https://www.youtube.com/watch?v=HW2Wcx-ktsc

4)https://www.youtube.com/watch?v=MYgmmJJfdBg

5) https://puniversity.informaticsglobal.com:2229/login.aspx

Topics relevant to “Skill Development”:

Page replacement algorithms, Scheduling policies, Deadlocks for Skill Development through Participative Learning techniques. This is attained through the assessment component mentioned in the course handout.